

Demo 1: Legacy Figure Reproduction

From a scanned geophysics figure to a reviewable parametric model

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Guided by prompts in iterations

Demo Idea

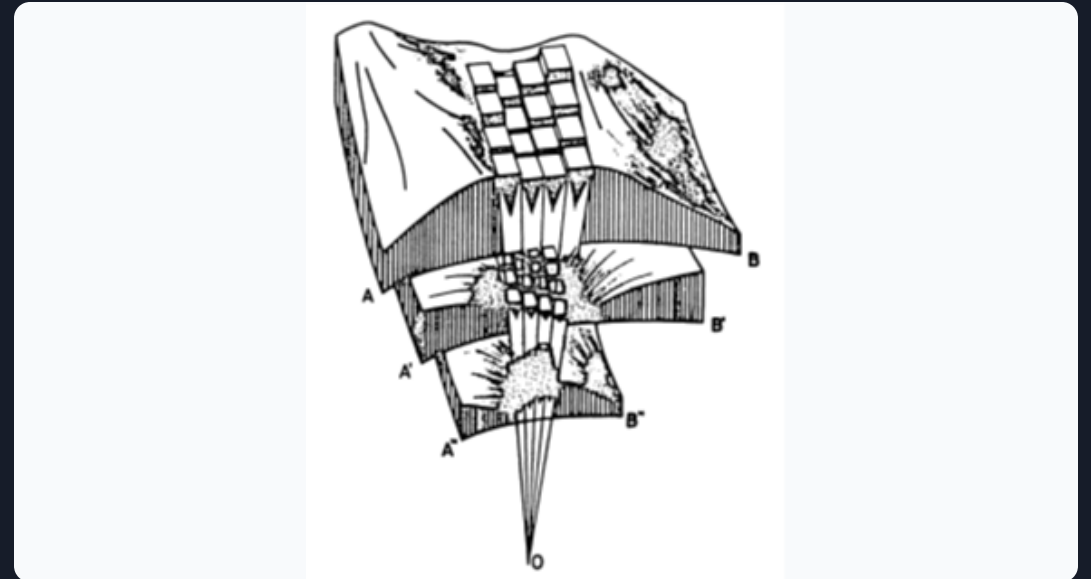
What we start from

- one scanned legacy figure in `input/figure.png`
- a task brief plus a detailed visual description
- no original plotting code

What learners practice

- turning a picture into explicit geometry
- iterating with focused prompts instead of one-shot requests
- keeping every draft image for comparison

Original target



Input Package

Files that frame the work

- `input/TASK.md`: reproduce the figure clearly and transparently
- `input/DATA.md`: explains slabs, prisms, apex, and labels
- `input/MILESTONES.md`: compare outputs and keep them numbered
- `prompts.md`: exact iteration prompts used during the demo

Deliverables

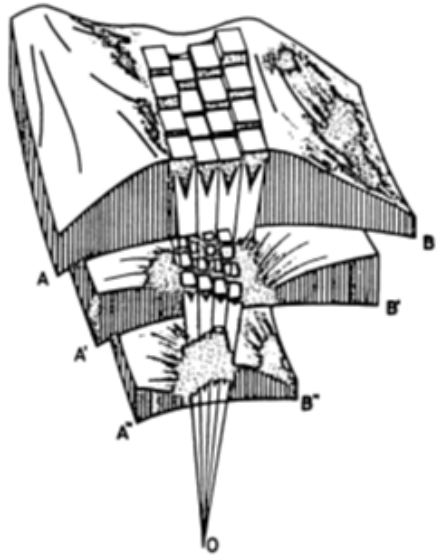
- code in `src/` that generates the figure
- numbered images in `output/`
- `requirements.txt` and `environment.yml`
- a `README.md` describing assumptions and adjustable parameters

Teaching point

The goal is not a perfect pixel-by-pixel redraw. The goal is a **reproducible geometric interpretation** that another researcher can inspect and improve.

Description of the figure

Original figure



Contents of `DATA.md`

The figure illustrates an early approach used in geophysics to model the Earth's subsurface by breaking it into a collection of simple, regularly shaped prism blocks. It represents the subsurface as a set of stacked geological layers, each shown as a broad, curved surface with vertical side walls. The vertical side walls are drawn with closely spaced parallel lines to indicate that each surface has thickness and represents a solid volume rather than just a flat boundary.

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Description of the task

Contents of `TASK.md`

Task

Reproduce the target figure from an older scientific paper in a transparent, reviewable way.

Technical approach

- Analyze the figure and its description, refer to `DATA.md` for details on the figure's content and context
- Use Python with libraries like Matplotlib/plotly/pyvista, NumPy, and possibly SciPy for numerical calculations.
- For reproducibility, prepare a conda environment file and a requirements.txt with all dependencies.
- Name the environment `legacy_figure_reproduction`
- Refer to `OUTPUTS.md` for the expected deliverables and structure.
- Use the `MILESTONES.md` as a step-by-step guide for the implementation process.

Description of the milestones

Contents of **MISSIONS.md**

1. Confirm the environment runs
2. Generate the code that produces the draft figure output
3. Compare the output to the target figure, identify discrepancies
4. Do not overwrite the generated figures, instead number them sequentially to track the progress and keep them for comparison
5. Iterate on the code to improve the figure, guided by the identified discrepancies
6. Refactor the code if needed for clarity
7. Document assumptions or approximations made in the process
8. Explore parameter choices that affect the output and document them

Description of the outputs

Contents of `OUTPUTS.md`

Required

- Reproduced figure image
- Code that generates the figure, organized in a clear way
- Detailed README with instructions on how to run the code and reproduce the figure, including any assumptions or approximations made, as well potential parameter choices that affect the output.

Output Structure

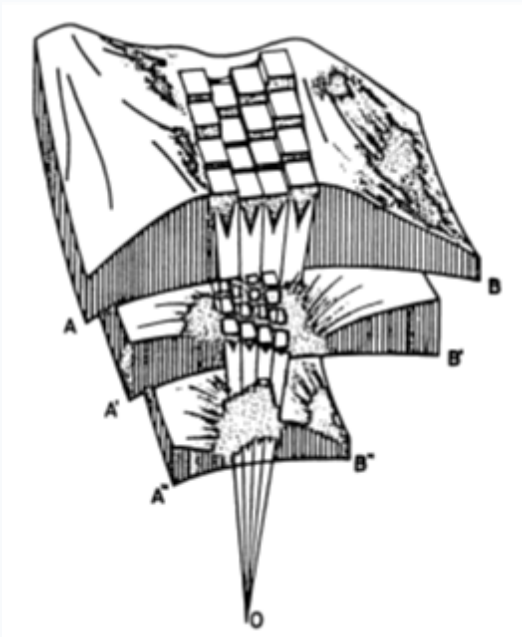
- use `src/` folder for code, including a clear entry point script that generates the figure
- add `requirements.txt` with all dependencies
- add `environment.yml` for conda environment setup
- use `output/` folder for the generated figure image

Iteration 1: First Runnable Approximation

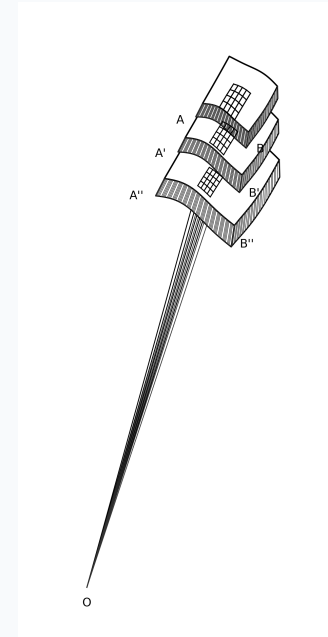
Prompt 1

Analyze the files in AI-Coding-Agents-Workshop-Instructors\demos\01-legacy-figure-reproduction\input folder.
Perform the task described in [TASK.md](AI-Coding-Agents-Workshop-Instructors/demos/01-legacy-figure-reproduction/input/TASK.md)

Original figure



Initial generated draft



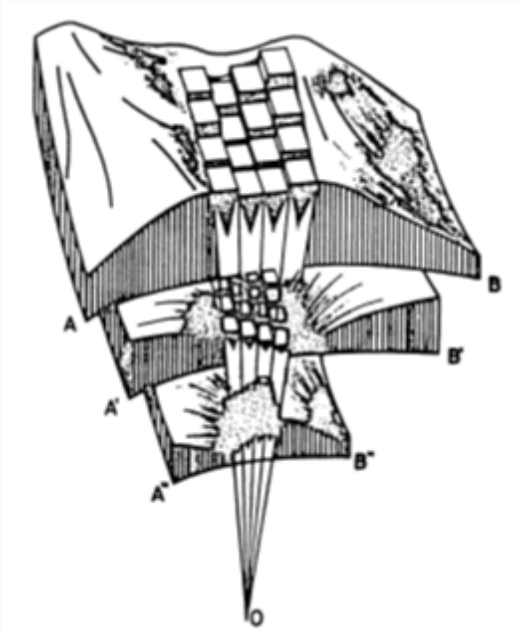
A useful first pass: slabs, grid, labels, and apex are already present, but the camera, slab proportions, and prism logic are still far from the target.

Iteration 1: Self-Critique and Self-Update

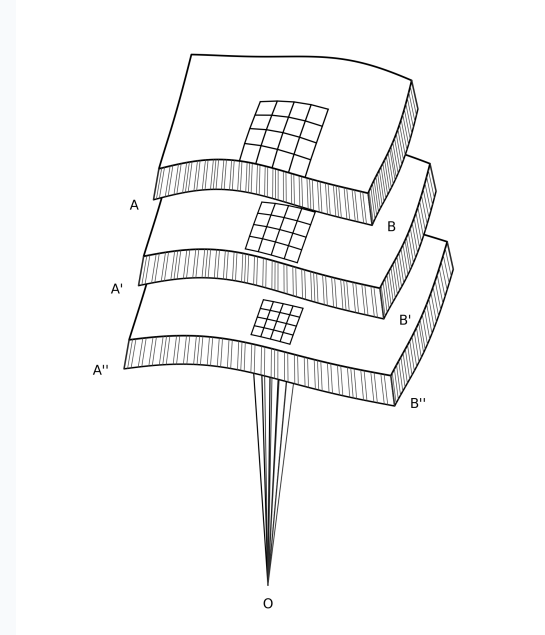
Prompt 1

Analyze the files in AI-Coding-Agents-Workshop-Instructors\demos\01-legacy-figure-reproduction\input folder.
Perform the task described in [TASK.md](AI-Coding-Agents-Workshop-Instructors/demos/01-legacy-figure-reproduction/input/TASK.md)

Original figure



Initial generated draft



Agent revised its approach based on the self-critique and updated the figure.

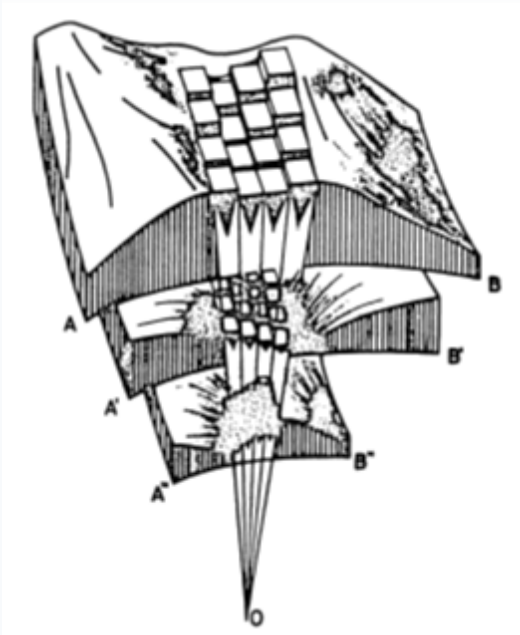
Iteration 4: Rebuild Around the 16 Prisms

Prompt 4

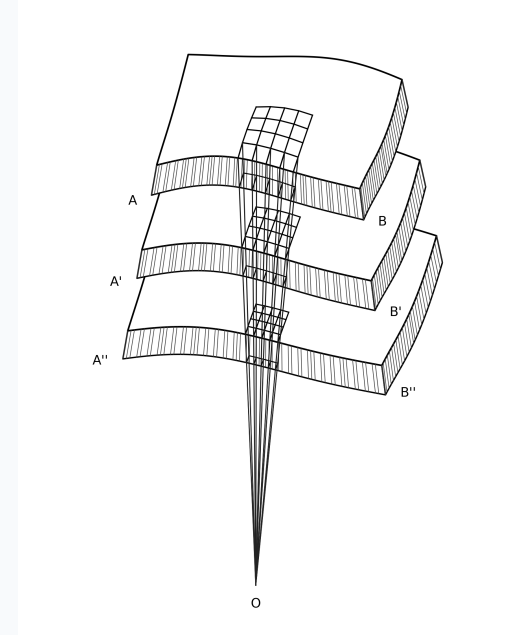
The result is not acceptable.

Thoroughly revise your plotting strategy. First plot the 16 prisms (4 x 4). Then add slab layers and make sure to intersect prisms with them properly, so that all the boundaries of the front 4 prisms are visible (although it is not like that in the original figure).

Original figure



After the plotting strategy reset



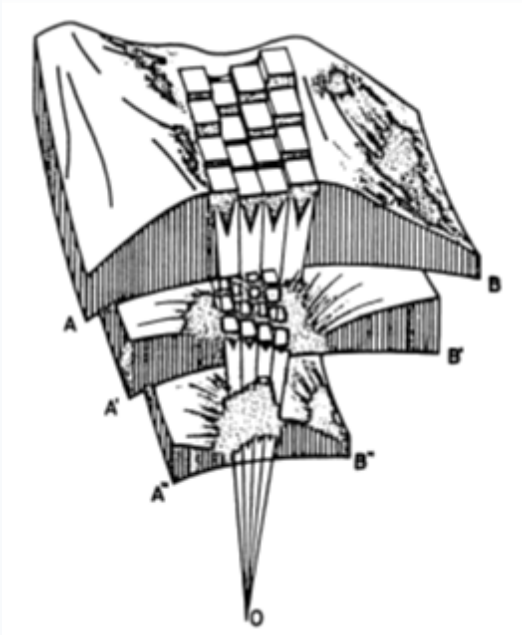
This is the first major conceptual jump: the figure is no longer a set of disconnected layer grids, but a prism bundle cutting through stacked slabs.

Iteration 8: Fix the View and Slab Footprints

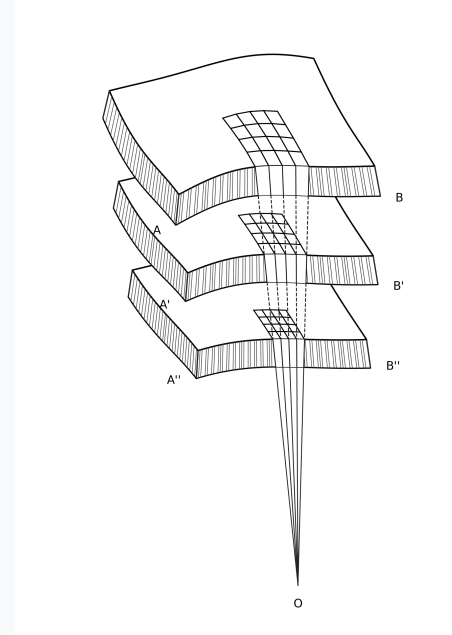
Prompt 8

Let's now play with the camera angle for the figure, to make it more like in the original one. In original the vertical walls are seen on the left sides of the slabs, in our case it is on the right side. So we need to turn the camera angle. Also, in original image, the upper slab is the largest, and the sizes of the slabs decrease with depth, opposite to the current figure.

Original figure



Camera and footprint correction



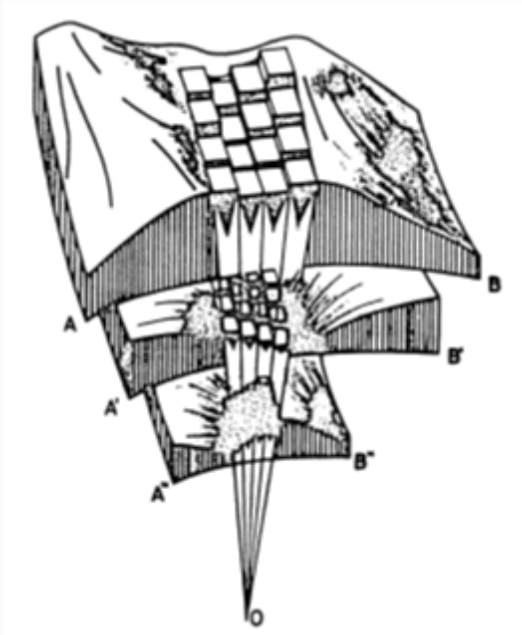
Now the left side walls are visible, and the slab sizes shrink with depth, which makes the overall silhouette much closer to the scanned source.

Iteration 15: Shape the Plateau Under the Grid

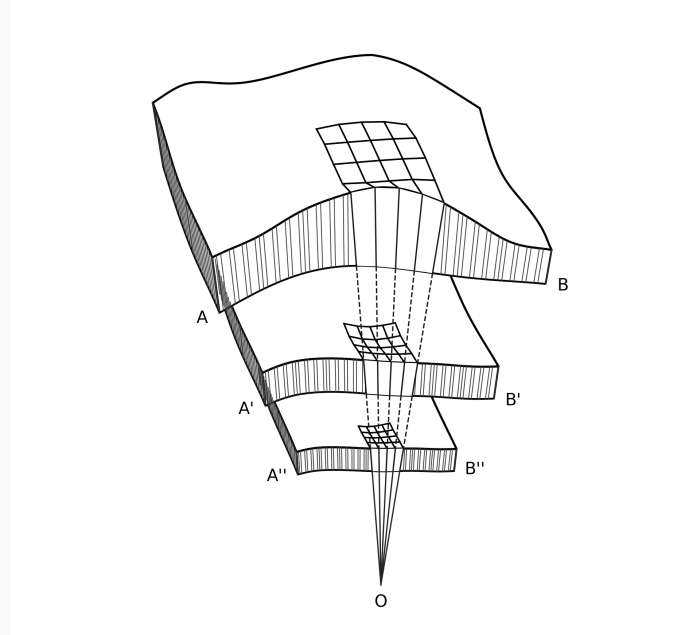
Prompt 15

Better. Now change the topography of the top slab such that there is a flat plateau in the part where the grid is placed. Also, increase a bit the width of the top slab, and reduce a bit the width of the bottom slab to make it look more like in original figure.

Original figure



Topography and relative slab sizes tuned



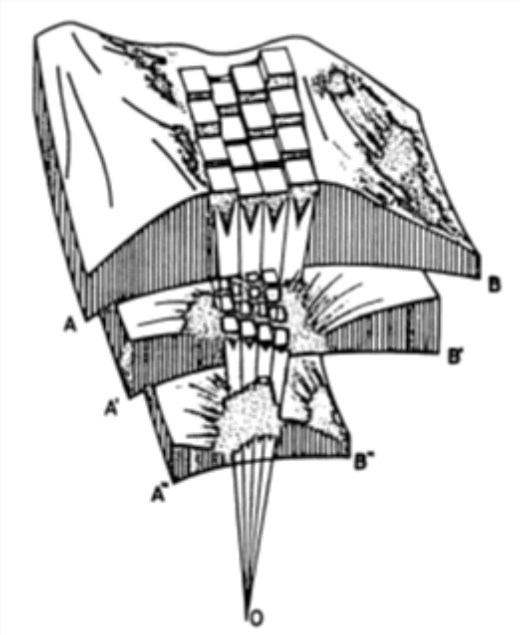
This iteration makes the grid readable again by flattening the local top surface while keeping the broader arch and asymmetry of the top slab.

Iteration 21: Redefine the Top Grid

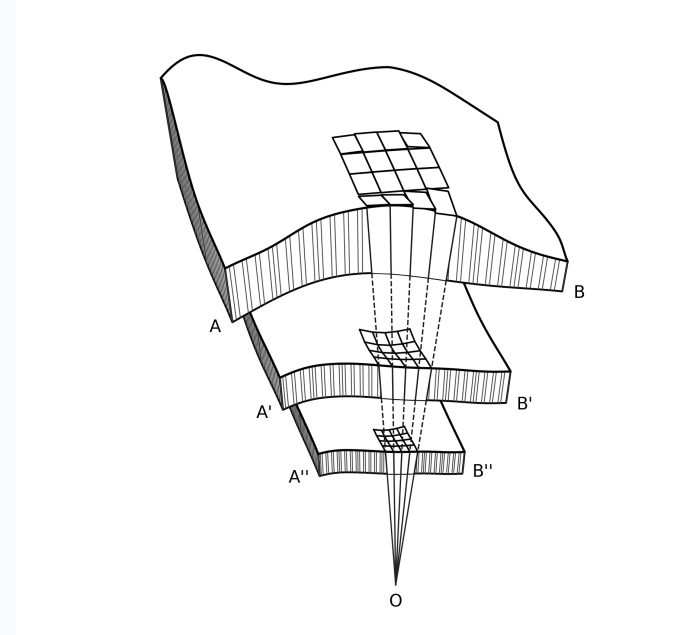
Prompt 21

Let's thoroughly revise how the grid is made. Instead of plotting the grid on the surface, we will plot the result of intersections of the prisms at different levels by a plane such that all these sections are co-planar. So these planes will be different for different prisms, like if trying to approximate the surface curvature by steps - planes making the prism bases. Let's do it for now for the top surface only.

Original figure



Top grid becomes stepped prism sections



Another key modeling shift: the top patch is no longer just drawn on the surface. It becomes a stepped approximation derived from prism-plane intersections.

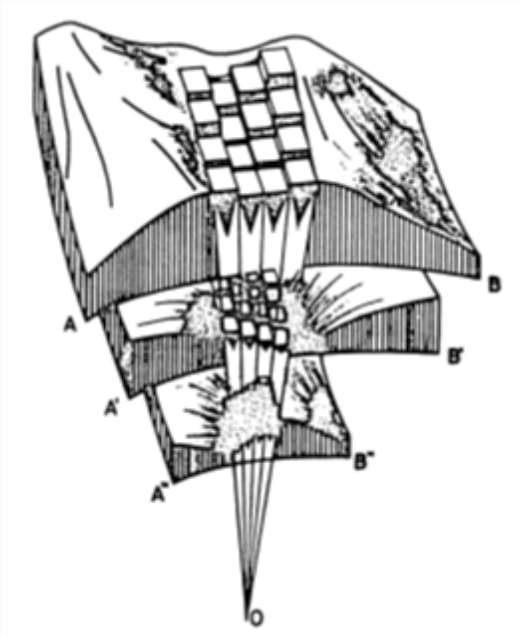
Iteration 25: Expose the Intrusion Zone Better

Prompt 25

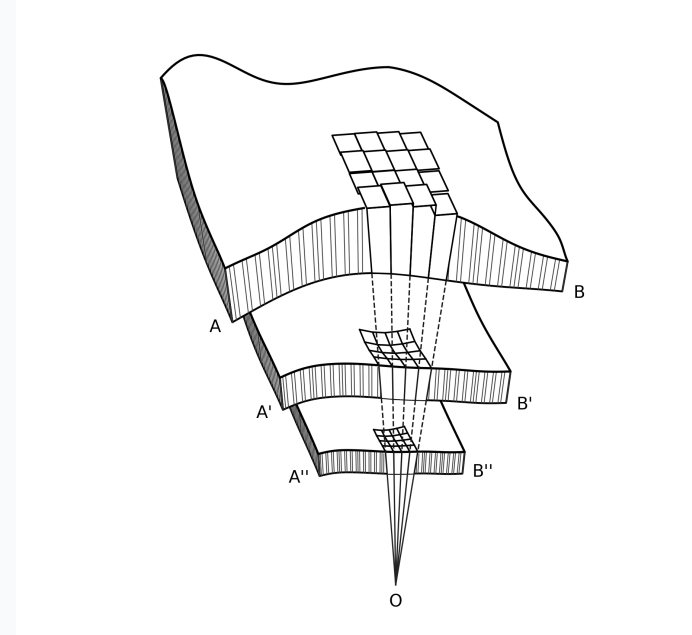
Beautiful!

Now let us a bit make the zone of the prism intrusion more variable, curve a bit, so that we can see more the different levels of the prism planes. like it is in the original image!

Original figure



Curved intrusion zone and clearer stepped levels



By curving the intrusion boundary, the stepped prism levels become easier to read, which makes the block-model idea much more explicit.

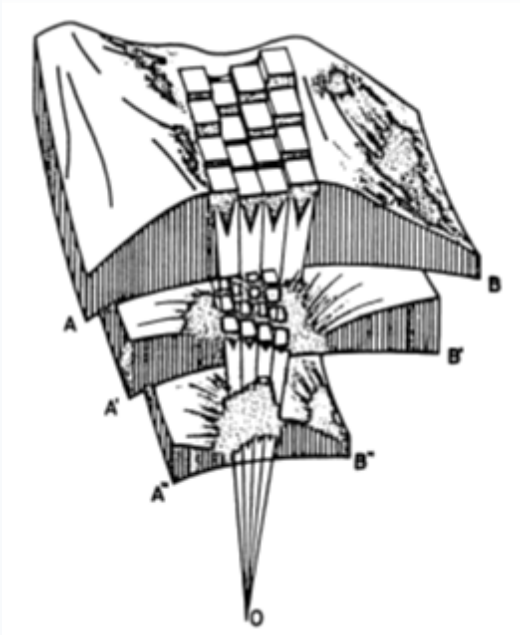
Final Iteration: Add Shading Without Hiding Structure

Prompt 27

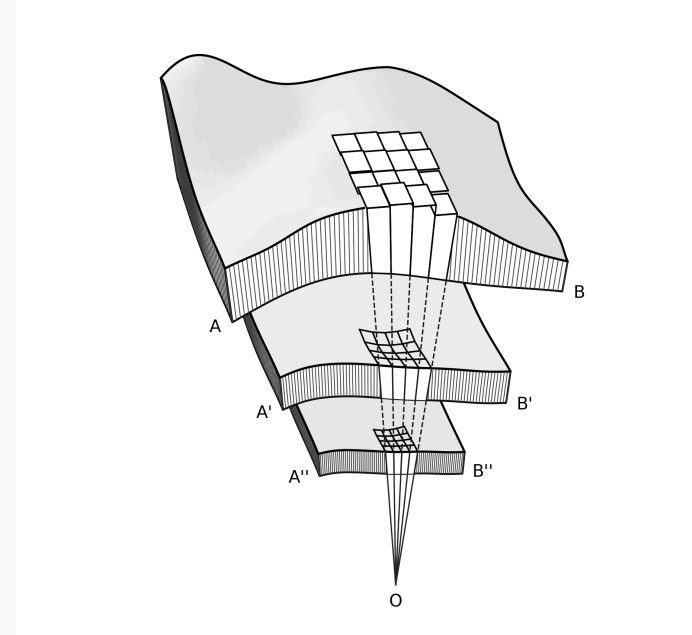
Perfect!

Now let us add a bit of an optional artistic touch. I want to shade the relief of the top surface like if there is a distant light source, from the left side. the lower surfaces should be just bit more gray to show it is shaded. Add this as a shading=true option on by default.

Original figure



Final shaded result



The final prompt adds relief shading as a parameterized visual layer. The geometry stays explicit and reviewable, but the figure reads more like the historical source.

Conclusions

- The hardest part was not drawing the lines; it was deciding which geometry the historical drawing implied.
- Short corrective prompts worked better than asking for a perfect redraw in one shot.
- Numbered outputs turned visual trial-and-error into something teachable and reviewable.
- The final result is still an approximation, but it is now reproducible, parameterized, and easy to discuss in terms of assumptions.